Treatments for age-related infertility are being developed at last

Startups hope to improve IVF by revealing the biology of infertility

FOR MANY well-known reasons, women around the world are increasingly choosing to have babies in their mid-30s and into their 40s. They face a considerable challenge: genetic errors in their eggs increase as they get older, leading to fewer pregnancies and more miscarriages.

In-vitro fertilisation (IVF) has long been the only solution offered to these women but, as it stands today, it is not fit-for-purpose and can be an expensive and heartbreaking gamble. If they can tolerate the financial, physical and emotional burden, would-be mothers are advised to try the process as many times as they can bear.

Change may be on its way. Two European startups want to increase women’ s chances of having a baby with treatments to improve the viability of human eggs. They are building on research that has peered deep inside the human egg to explain exactly why baby-making becomes so error-prone as women age. By identifying the molecules and processes responsible for faulty eggs, the companies hope to improve IVF outcomes for older women.

A woman’ s oocytes (immature eggs) are some of the oldest cells in her body. They were formed when she was a fetus; her ovarian reserve peaked in the millions while she was still in her mother’ s womb. By puberty, the stock has plummeted to 300,000-400,000. She will continue to lose oocytes every month up until reserves are virtually depleted at menopause.

It is not only the quantity of oocytes that drops with time: so does their quality. Older ones are more likely to carry chromosomes in abnormal numbers. Healthy human cells have 23 pairs of matching chromosomes. As egg cells develop, each pair is joined and lined up carefully, ensuring that they can be split such that half—one complete set—are discarded. This leaves mature eggs with 23 single chromosomes that can partner with the corresponding number in a sperm. If something goes wrong in the process, an “aneuploid” egg can be formed: one that has either too few or too many chromosomes.

Aneuploidy is a leading cause of infertility. In women in their 20s over 20% of oocytes have an incorrect number of chromosomes. The share rises to 50% in those over the age of 40. No one has yet developed a solution to the problem.

Around the beginning of the decade a team led by Melina Schuh of the Max Planck Institute for Multidisciplinary Sciences in Göttingen, Germany, began mapping out the

molecular causes of aneuploidy. They showed that levels of cohesin, a molecule which holds chromosome pairs together as an egg develops, decline with age. Diminished cohesin disrupts the chromosomal choreography and can result in aneuploid eggs. The team also found that a protein called Shugoshin 1 (derived from a Japanese word meaning "guardian spirit") plays a crucial role in preventing cohesin's decay, and thus lowers the risk of oocytes turning into aneuploid eggs.

In January 2025 Dr Schuh and her collaborator Agata Zielinska founded a company, Ovo Labs, to turn these discoveries into treatments. Ovo Labs is developing a product called EmbryoProtect 1, based on Shugoshin 1, that can be injected into immature egg cells before aneuploidy develops. In lab experiments the team injected EmbryoProtect 1 first into mouse eggs and later into more than 100 human eggs donated by IVF patients.

Early results appear promising. "When a mouse gets older, 60% of the eggs become chromosomally abnormal," says Dr Zielinska. "But when you treat an old mouse egg with EmbryoProtect 1 you can see that you almost fully eliminate the chromosomal defects." A first generation of healthy mouse pups has been born by fertilising eggs treated with EmbryoProtect 1. When the team injected oocytes with the product, they found that the proportion that developed chromosomal errors fell by roughly half compared with an untreated control group.

Dr Schuh says data from Ovo Labs and her laboratory at the Max Planck Institute show that in oocytes from women aged 34-40, the proportion of chromosomally normal eggs more than doubled after treatment. Dr Zielinska and her collaborator Debojit Saha, now at Princeton University, presented these results, which are not yet peer reviewed, at the annual meeting of the American Society for Reproductive Medicine, in Baltimore, on October 26th and October 28th. If their findings translate into clinical practice, women undergoing IVF could have "substantially more" chromosomally normal eggs available for their IVF cycles, says Dr Schuh.

A second startup, U-Ploid Biotechnologies based in Oxford, was created by Alexandre Webster and Chloe Charalambous, two former members of Dr Schuh's lab. Instead of replicating Shugoshin 1, the firm has developed a molecule called Lyvanta that slows down separase, an enzyme which cleaves Shugoshin 1, leading to less cohesin and, eventually, more aneuploidy.

In experiments on mice publicised in 2025, Lyvanta reduced the incidence of chromosomal defects in 84% of eggs. U-Ploid is now testing their molecule in human cells. In preliminary results also presented in Baltimore, it said that treating immature eggs with Lyvanta had almost doubled the number which matured to be chromosomally normal. The team said that 66% of cells treated with Lyvanta did not have any chromosome errors, compared with 36% of untreated eggs.

Ovo Labs is exploring two further approaches to reduce aneuploidy. One is an engineered cohesin that would help its natural counterpart hold chromosomes together as eggs mature. The first results are expected to be published in a scientific journal on November 5th.

The other approach builds on an understanding of how immature human eggs shed their excess chromosomes. That process involves microscopic spindles which line up the chromosomes inside the egg as it divides. Dr Schuh's research has shown that this spindle structure is particularly unstable in human eggs, even in younger women, in comparison with other mammals. This could explain why one in five eggs from women in their 20s is aneuploid, as well as why miscarriages are common even in younger women.

Comparisons with mice, cows and pigs revealed that human oocytes have very low levels of a spindle-stabilising protein called KIFC-1. Injecting KIFC-1 back into human egg cells reduced the risk of aneuploidy. Dr Schuh says Ovo Labs is exploring this avenue in the hopes that boosting spindle stability could improve IVF outcomes for both younger and older patients.

Rita Vassena, a fertility specialist and industry veteran at CooperSurgical, an American health-care firm based in Connecticut, calls preclinical data from both U-Ploid and OvoLabs "encouraging" but cautions that the technologies "will need to prove their short- and long-term safety and efficacy". Still, says Dr Vassena, "Fixing the problem at the source brings about an unprecedented potential for treatment-efficiency gains."

Demand will be high if any of these approaches pass clinical testing. In America more than 40% of women seeking fertility treatments were over 38 in 2018. In Britain, one in six patients starting IVF in 2023 was aged 40 and above. The researchers at Ovo Labs and U-Ploid say they frequently have to turn away women wanting to get in early on clinical trials that do not yet exist.